

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Numerical Method**

Code No: **CSC212** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. How secant methods differs from Newton Raphson method? Derive the formula for Secant Method. Solve the equation $\cos x + 2 \sin x - x^2 = 0$ using Secant method. Assume error precision as 0.01. Discuss drawbacks of the Newton Raphson method.
2. Define the terms interpolation and extrapolation. Write down algorithm and program for Newton's divided difference interpolation.
3. How Gauss Jordan method differs from Gauss Elimination method? Solve the following system of equations using Gauss Jordan method. How can we use Gauss Jordan method to find inverse of a matrix? Discuss.
$$\begin{matrix} 2x + 4z = 15 \\ 2x + 3y - 2z = 43 \\ x + 2y - 4z = -4 \end{matrix}$$

Section B

Attempt any Eight questions[8×5=40]

4. Define the terms approximate error and relative approximate error? Discuss the working of Half Interval method for finding roots of non-linear equation.
5. Construct Newton's backward difference table for the given data points and approximate the value of $f(x)$ at $x=45$.

x	10	20	30	40	50
f(x)	0.9850	0.9340	0.8660	0.7660	0.643
6. Fit the quadratic curve through the following data points and estimate the value of $f(x)$ at $x=2$.

x	1	3	5	7
y	2.1345	6.278	7.5	9
7. Factorize the following matrix using Cholesky method.
$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 9 \end{bmatrix}$$
8. How can we calculate derivatives of discrete (tabulated) functions? Write down its algorithm.
9. Find following integral using composite trapezoidal rule for using 2 segments ($k=2$) and 4 segments ($k=4$).
$$\int_0^1 (x^3 + 2) dx$$
10. Approximate the solution of $y' = 3x^2$, $y(1) = 1$ using Taylor's Series method using first four terms. Approximate the value of $y(2)$.
11. Solve the Poisson's equation $\nabla^2 f = xy$ with $f = 2$ on boundary by assuming square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ and $h = 1$.

12. Write down program for solving ordinary differential equation using Heun's method.